

The second artifact selected comes from CS 360: Mobile Architecture and Programming, where I developed a mobile application designed to track user weight and manage personal health data. The application includes user interface components, input handling, and data management using a SQLite database. This artifact demonstrates my ability to build user centered applications while integrating backend data processing within a mobile environment.

I selected this artifact because it highlights my ability to apply algorithmic thinking and data structure concepts to improve system performance and usability. While the original implementation focused on functionality, the enhancements allowed me to refine how data is processed, retrieved, and displayed within the application.

To enhance this artifact, I improved the efficiency of data retrieval by implementing sorting and filtering techniques within database queries. I modified the application to retrieve weight data in descending order by date, ensuring that the most recent data is displayed first. This not only improves usability but also demonstrates the application of sorting algorithms within a real world system. Additionally, I implemented a filtering mechanism using SQL LIKE queries, allowing users to search for specific records based on defined criteria. This reduces unnecessary data processing and improves application performance.

These enhancements demonstrate the importance of selecting appropriate data structures and optimizing algorithms to improve system efficiency. By using structured queries and efficient data handling techniques, I minimized redundant operations and improved the responsiveness of the application. This reflects a strong understanding of how algorithmic decisions impact real world application performance.

From a security perspective, I considered the importance of controlled data access and input handling within the application. By using structured queries and limiting the scope of returned data, the application reduces the risk of improper data exposure and improves overall data integrity. This demonstrates an awareness of how algorithm design and data handling practices can contribute to secure application development.

From a stakeholder perspective, these improvements enhance the user experience by providing faster access to relevant data and more intuitive interaction with the application. Efficient data processing supports better decision making for users tracking their health data, while also improving the overall usability of the system.

This artifact aligns with program outcomes by demonstrating the ability to design and evaluate computing solutions using algorithmic principles and data structures. Through this enhancement, I strengthened my ability to apply theoretical concepts to practical applications and gained a deeper understanding of how small improvements in logic can significantly impact performance and user experience.